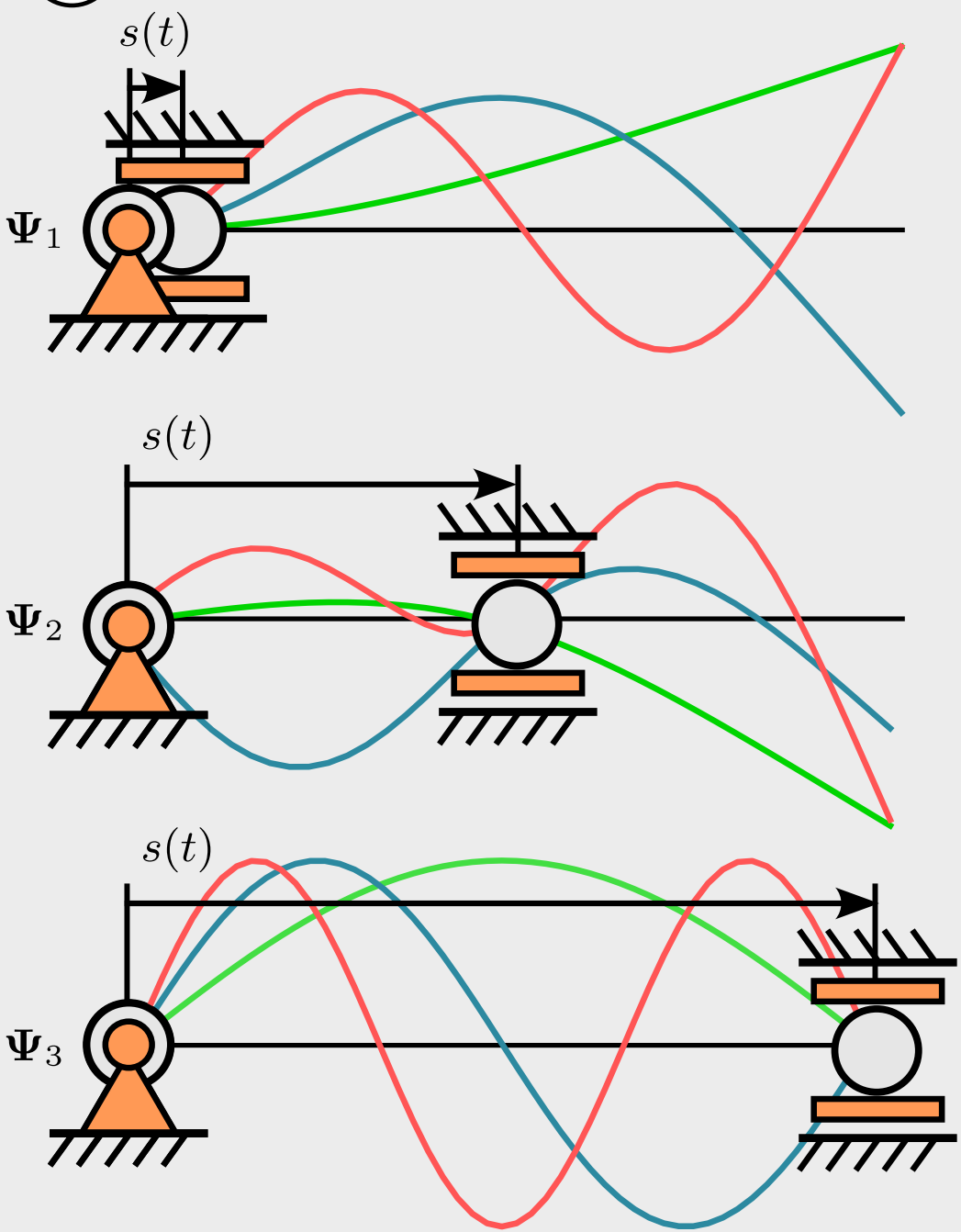
1

collect snapshots, e.g.:

$n_{\text{snap}} = 3$ and $r_{\text{snap}} = 3$



2

assemble *pool*

$$\Psi_{\text{all}} = [\Psi_1, \Psi_2, \Psi_3]$$

$$\Psi_{\text{all}} \in \mathbb{R}^{n_{\text{DOF}} \times (n_{\text{snap}}\, r_{\text{snap}})}$$

3

perform POD

$$\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{\text{T}} = \text{SVD}\left(\Psi_{\text{all}}\right)$$

4

select global modes

$$\Psi_{\text{glob}} = \boldsymbol{U}_{:,1:r_{\text{glob}}}$$

$$\Psi_{\text{glob}} \in \mathbb{R}^{n_{\text{DOF}} \times r_{\text{glob}}}$$

5

reduce system (const. matrices)

$$\boldsymbol{M}_{\text{red}} = \Psi_{\text{glob}}^{\text{T}}\, \boldsymbol{M}\, \Psi_{\text{glob}}$$

$$\boldsymbol{K}_{\text{red}} = \Psi_{\text{glob}}^{\text{T}}\, \boldsymbol{K}\, \Psi_{\text{glob}}$$

6

set up Lagrangians Type 1

$$\boldsymbol{f}_{\text{red}}(t) = \Psi_{\text{glob}}^{\text{T}}\, \boldsymbol{f}(t)$$

$$\boldsymbol{C}_{q,\text{red}}(s(t)) = \boldsymbol{C}_q\left(s(t)\right) \Psi_{\text{glob}}$$

$$\boldsymbol{M}_{\text{red}}\, \ddot{\boldsymbol{p}}(t) + \boldsymbol{K}_{\text{red}}\, \boldsymbol{p}(t) + \boldsymbol{C}_{\text{q,red}}^{\text{T}}\left(s(t)\right) \boldsymbol{\lambda}(t) = \boldsymbol{f}_{\text{red}}(t)$$

$$\boldsymbol{g}\left(s(t), \Psi_{\text{glob}}\, \boldsymbol{p}(t)\right) = \boldsymbol{0}$$